An interval-length boundary for the open spectral-set number-rigidity question

Candidate substantial partial answer to the question of arXiv:1702.07323

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Status: candidate partial result, likely valid. For a widely separated countable union of intervals, we give an exact interval-length criterion for *linear* number rigidity. Exponentially decaying interval lengths produce genuinely number-rigid open spectral sets with infinitely many components. Polynomially decaying lengths produce explicit open finite-measure sets whose DPPs are not linearly number rigid. Whether the latter are nonlinearly number rigid remains open.

Source question

On PDF page 3 of *Rigid stationary determinantal processes in non-Archimedean fields*, Qiu asks [1]:

> Let $S \subset \mathbb{R}$ be an open subset with finite positive Lebesgue measure. Is the determinantal point process on $\mathbb{R}$ induced by the correlation kernel $\widehat{\mathbf{1}_S}(x-y)$ number rigid?

The complete source page is reproduced below.

we know that the kernel

$$p^{-n}\mathbb{1}_{p^{-n}\mathbb{Z}_p}(x-y) = \widehat{\mathbb{1}_{p^n\mathbb{Z}_p}}(x-y), \quad (x, y \in \mathbb{Q}_p) \tag{1.3}$$

induces a determinantal point process on $\mathbb{Q}_p$. This determinantal point process can be trivially verified to be number rigid since for any pair $x, y \in \mathbb{Q}_p$ such that $|x-y|_p > p^n$, we have

$$\widehat{\mathbb{1}_{p^n\mathbb{Z}_p}}(x-y) = 0$$

and this implies that the determinantal point process induced by the kernel (1.3) is the union of countably many independent copies of determinantal point processes on the translates of $p^{-n}\mathbb{Z}_p$, each of them has exactly one particle. By the same reason, if $S \subset F$ is a **union of finitely many balls**, then the determinantal point process induced by the kernel $\widehat{\mathbb{1}_S}(x-y)$ is trivially number rigid. Of course, in these trivial examples, the sets S satisfy trivially the condition (1.1).

Theorem 1.2 produces non-trivial examples of Borel subsets $S \subset F$ such that the associated stationary determinantal point processes are number rigid, see §4.2 for such examples. Since any finite dimensional vector space over F can be seen as a finite extension of the field F, which is again a non-Archimedean local field, Theorem 1.2 produces non-trivial examples of stationary determinantal point processes on any finite dimensional vector space over F.

Recall that any open set of the real line $\mathbb{R}$ is a countable union of intervals, inspired by our result in non-Archimedean situation, it is natural to ask the following

Question. *Let $S \subset \mathbb{R}$ be an* **open** *subset with finite positive Lebesgue measure. Is the determinantal point process on $\mathbb{R}$ induced by the correlation kernel*

$$\widehat{\mathbb{1}_S}(x-y), \quad (x, y \in \mathbb{R})$$

number rigid?

2. Preliminaries

2.1. **Notation.** We recall some basic notion in the theory of local fields. Let F be a non-discrete non-Archimedean local field. The classification of local fields (see Ramakrishnan and Valenza[12, Theorem 4-12]) implies that F is isomorphic to one of the following fields:

- a finite extension of the field $\mathbb{Q}_p$ of p-adic numbers for some prime p.
- the field of formal Laurent series over a finite field.

Let $|\cdot|$ denote the *absolute value* on F and let $d(\cdot,\cdot)$ denote the metric on F defined by $d(x, y) = |x-y|$. The set $\mathcal{O}_F = \{x \in F : |x| \leq 1\}$ forms a subring of F and is called the ring of integers or the valuation ring of F. The subset $\mathcal{M}_F = \{x \in F : |x| < 1\}$ is the unique maximal

The question is universal: it asks about every such open set. Our result settles a broad positive subfamily and identifies an explicit family for which all L^2 linear-statistic methods necessarily fail.

The structure factor

For a measurable $S \subset \mathbb{R}$ of finite positive measure, let $\mathcal{X}_S$ be the stationary projection DPP with kernel

$$K_S(x, y) = \widehat{\mathbf{1}_S}(x - y).$$

Write $\rho = |S|$. Up to Fourier-normalization constants and the harmless rescaling to unit intensity, the continuous covariance spectral density, or structure factor, is

$$s_S(t) = \rho - |S \cap (S + t)| = \frac{1}{2}|S \triangle (S + t)|. \tag{1}$$

Indeed, Plancherel and the convolution theorem give $|\widehat{K_S(0, \cdot)}|^2(t) = |S \cap (S + t)|$. Multiplying s_S by a positive constant does not affect any reciprocal-integrability statement.

We use the precise conclusion that is safe in the later criterion of Lachièze-Rey [2]: in dimension one, a stationary DPP is *linearly* number rigid on bounded intervals exactly when

$$\int_0^{\varepsilon} \frac{dt}{s_S(t)} = \infty. \tag{2}$$

See PDF page 19, especially the sentence following Corollary 2. The paper explicitly cautions on PDF page 8 that the word "linear" is sometimes omitted but all its methods concern linear rigidity. Divergence in (2) is nevertheless enough for ordinary Ghosh–Peres number rigidity, since L^2 linear reconstruction is an actual measurable reconstruction. Convergence proves only failure of linear number rigidity.

Exact classification for separated intervals

Let $(\ell_n)_{n \geq 1}$ satisfy

$$0 < \ell_n \leq 1, \qquad \sum_{n \geq 1} \ell_n < \infty,$$

and define the open set

$$S_\ell = \bigcup_{n \geq 1} (3n, 3n + \ell_n). \tag{3}$$

Theorem 1. *For* $0 < |t| < 1$*,*

$$s_{S_\ell}(t) = \omega_\ell(|t|), \qquad \omega_\ell(u) := \sum_{n \geq 1} \min\{u, \ell_n\}. \tag{4}$$

Consequently $\mathcal{X}_{S_\ell}$ *is linearly number rigid if and only if*

$$\int_0^{\varepsilon} \frac{du}{\sum_n \min\{u, \ell_n\}} = \infty. \tag{5}$$

Whenever (5) *holds,* $\mathcal{X}_{S_\ell}$ *is also number rigid in the ordinary nonlinear sense.*

Proof. For $|t| < 1$, an interval in (3) can meet only its own translate; distinct three-unit blocks remain disjoint. The overlap contributed by the nth interval is $(\ell_n - |t|)_+$. Thus

$$|S_\ell| - |S_\ell \cap (S_\ell + t)| = \sum_n \bigl(\ell_n - (\ell_n - |t|)_+\bigr) = \sum_n \min\{|t|, \ell_n\},$$

which proves (4). The equivalence is (2). If the integral diverges, linear reconstruction gives number rigidity for bounded intervals. For an arbitrary bounded Borel set B, choose a bounded interval $I \supset B$: the outside of B reveals $\mathcal{X}(I \setminus B)$, while the outside of I determines $\mathcal{X}(I)$, so $\mathcal{X}(B) = \mathcal{X}(I) - \mathcal{X}(I \setminus B)$ is also determined. $\square$

This formula is also the exact competition between the number of intervals longer than the translation scale and the total mass of shorter intervals:

$$\omega_\ell(u) = u\,\#\{n : \ell_n \geq u\} + \sum_{\ell_n < u} \ell_n.$$

Two opposite infinite-component regimes

Corollary 2 (A genuinely rigid open set)**.** *For $\ell_n = 2^{-n}$, the open set S_ℓ has finite positive measure, infinitely many components, and $\mathcal{X}_{S_\ell}$ is number rigid.*

Proof. If $2^{-(N+1)} < u \leq 2^{-N}$, then

$$\omega_\ell(u) = Nu + \sum_{n > N} 2^{-n} = Nu + 2^{-N} \asymp u \log(1/u).$$

Hence (5) diverges like $\int_0 du/(u \log(1/u))$. Theorem 1 applies. $\square$

This supplies a particularly elementary positive infinite-union subcase. Lin, Qiu, and Wang independently constructed more intricate rigid generalized Cantor sets and complementary infinite unions [3]; their PDF page 1 states the prior finite-union boundary and the goal of finding special infinite-union examples.

Corollary 3 (A sharp linear obstruction family)**.** *Fix $p > 1$ and put $\ell_n = n^{-p}$. Then S_ℓ is open and has finite positive measure, but $\mathcal{X}_{S_\ell}$ is not linearly number rigid. More precisely,*

$$s_{S_\ell}(t) \asymp |t|^{1-1/p} \qquad (t \to 0).$$

In particular, for $p = 2$ one obtains the explicit set

$$S = \bigcup_{n \geq 1} (3n, 3n + n^{-2}), \qquad s_S(t) \asymp \sqrt{|t|}.$$

Proof. Take $N = \lfloor u^{-1/p} \rfloor$. Then

$$\omega_\ell(u) = Nu + \sum_{n > N} n^{-p}.$$

Both terms are comparable to $u^{1-1/p}$, by the integral bounds for the p-series tail. Since $1 - 1/p < 1$, the reciprocal is locally integrable. The equivalence in Theorem 1 proves failure of linear number rigidity. $\square$

Deep upgrade attempt and exact obstruction

The disjoint frequency intervals give an orthogonal decomposition of the Fourier projection. A tempting upgrade is therefore to regard $\mathcal{X}_{S_\ell}$ as an unlabelled superposition of independent sine processes and study mark erasure. This route fails at its first probabilistic step: orthogonal sum of projection *operators* does not make the corresponding spatial DPP the independent union of the component DPPs. Expanding the determinant of the summed kernel leaves mixed interference terms; independence occurs only in special spatially block-diagonal situations.

A direct Palm/deletion-tolerance route also stops at a real obstruction. The reduced Palm kernel is a rank-one modification of the full projection, but neither the interval-length formula nor the second-order structure factor controls absolute continuity of its point-process law. Thus convergence of (5) cannot honestly be promoted to nonlinear non-rigidity. This is not a cosmetic gap: arXiv:2409.18519 explicitly separates linear from nonlinear rigidity, and known perturbed lattices show that nonlinear rigidity can survive failure of linear rigidity.

Accordingly, Corollary 3 is a strong method barrier, not a full counterexample to Qiu's question. Settling it would require a new theorem showing deletion tolerance, equivalence of different-order Palm laws, or an independent nonlinear reconstruction mechanism.

Novelty and verification bounds

Searches through 11 August 2026 covered the exact question, arbitrary open spectral sets, infinite unions of intervals, structure-factor criteria, superpositions of sine processes, Palm equivalence, and deletion tolerance. No located source states Theorem 1 or the polynomial obstruction family. The decisive analytic criterion is from arXiv:2409.18519, so novelty is claimed only for the interval-length reduction, its phase boundary, and the explicit application to the source question.

The accompanying script checks the exact finite-truncation overlap formula and the predicted exponential and polynomial asymptotics. It is a sanity check, not part of the proof.

References

[1] Y. Qiu, *Rigid stationary determinantal processes in non-Archimedean fields*, arXiv:1702.07323, 2017. Question on PDF p. 3.

[2] R. Lachièze-Rey, *Rigidity of random stationary measures and applications to point processes*, arXiv:2409.18519v3, 2025. Linear-rigidity caveat on PDF p. 8; stationary-DPP criterion and linear conclusion on PDF p. 19.

[3] Z. Lin, Y. Qiu, and K. Wang, *Number rigid determinantal point processes induced by generalized Cantor sets*, arXiv:2407.14168, 2024.